PAGE 1 · THE STRATEGIC IMPERATIVE

Why Banks Need PeX

Payment exceptions are a resolution problem — not simply a transaction-processing problem.

PAYMENT EXCEPTION LANDSCAPE: FRAGMENTATION & SILOS

SWIFT MT / MX
ISO 20022 & CBPR+

Fedwire / CHAPS
High-Value RTGS

ACH / NACHA / SEPA
Batch & Direct Debit

Instant / RTP
ISO Real-Time

PAYMENT EXCEPTION OCCURS (Breaks · Rejections · Repair Queues)
41 Canonical Break Types across Account, Syntax, Sanctions, Liquidity, and Duplicates

Payment Hub
ACI / Finastra / FIS

Core Banking
DDA Ledgers

Sanctions / AML
Fircosoft / Actimize

KYC / Directory
Client Master

Clearing Rails
SWIFT / Fedline

SME Knowledge Silos
70%+ manual Excel & local macros

Manual System Lookups
6–10 disconnected queries per break

Asynchronous Handoffs
Email & ticket delays inflate cycle time

RESULT: Long cycle times, rising SLA penalties, inconsistent repair quality, and trapped operational knowledge.

THE STRUCTURAL GAP & THE NEED

1. Transaction Processing
Hubs process happy-path payments linearly

2. Exception Occurs
Breaks span multiple rails, systems & policies

3. Who Resolves?
No single system owns cross-silo context

4. Governed Intelligence Needed
Common layer between breaks & bank execution

EXECUTIVE REALITY
"Payment engines process transactions.
They do not inherently provide contextual resolution intelligence when execution breaks."

OPERATIONAL MATURITY ► PEOPLE-DRIVEN (Manual & SME Silos) ► PROCESS-DRIVEN (Task Automation) ► WORKFLOW-DRIVEN (Ticketing & Queues) ► RESOLUTION-INTELLIGENCE (Governed PeX Platform)

CORE TAKEAWAY: Banks need more than case management or workflow. They need structured resolution intelligence for payment exceptions.

What Is PeX?

A structured Payment Exception Resolution intelligence layer — built around 41 exception profiles.

THE 41 CANONICAL EXCEPTION PROFILES UNIVERSE

41 CANONICAL EXCEPTION PROFILES
Complete BIAN-Aligned Payment Break Taxonomy

Account & Routing (12 Profiles)
Invalid IBAN/BBAN · Closed DDA · Missing BIC · Routing Mismatch

Data & Formatting (9 Profiles)
ISO 20022 XML Syntax · MT103 Truncation · Charge Codes (OUR/BEN/SHA)

Compliance & Sanctions (8 Profiles)
OFAC 50% Disambiguation · PEP False-Positives · FATF Travel Rule

Liquidity & Timing (6 Profiles)
Clearing Cut-off Breaches · Debit Cap Limit · Value Date · Cover Delay

Duplicate & Processing (6 Profiles)
Duplicate UETR · pacs.004 Return Processing · camt.056 Recall Handling

COMMON SEMANTICS ➔ CANONICAL MODEL ➔ REUSABLE RESOLUTION PATTERNS
41 Canonical Profiles normalize break semantics across any bank core, payment hub, or clearing rail.

WHY PeX IS MORE THAN AI: GOVERNED VS GENERIC

GENERIC AI ALONE *Unconstrained & Heuristic*	PeX RESOLUTION PLATFORM *Deterministic & Governed*
Raw Prompt *Text input from user*	**41 Exception Profiles** *Canonical break classification*
Unbounded LLM *Probabilistic inference*	**Peripheral Evidence** *Live DDA, KYC, Sanctions ICDs*
Unstructured Text *Prone to hallucinations*	**AI ReasoningResult** *Typed Pydantic JSON advice*
No Banking Controls *No deterministic validation*	**Deterministic Rules** *100% banking rule precedence*
Opaque Output *No regulatory audit trail*	**Governed Outcome** *SHA-256 Merkle provenance*

CORE DISTINCTION: PeX is not an LLM sitting on payment data.
It is a structured resolution capability built around a defined exception universe.

CANONICAL RESOLUTION LAYER ▶ 41 EXCEPTION PROFILES ▶ BANKING ONTOLOGY ▶ CANONICAL MODEL ▶ PERIPHERAL EVIDENCE ▶ COMMON RESOLUTION INTELLIGENCE

CORE TAKEAWAY: PeX is not 'AI for payments.' It is a structured resolution capability built around a defined exception universe.

PAGE 3 · HERO LOGICAL ARCHITECTURE

How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

PAYMENT HUBS (ACI · Finastra · FIS)
Inbound Exception Feeds

CORE BANKING (DDA Ledgers · Accounts)
Posting Breaks & Balances

CLEARING RAILS (SWIFT · Fedwire · CHAPS)
Protocol Rejections & Returns

PeX GOVERNED RESOLUTION ENGINE (LOGICAL ARCHITECTURE)

1. INGEST ➔ NORMALIZE ➔ CLASSIFY
Canonical PaymentExceptionCase + 41-Profile Map

2. EVIDENCE & CONTEXT CORROBORATION
Read-Only ICDs: DDA · Sanctions · KYC · SWIFTRef

3. AI MODEL GATEWAY (ReasoningResult)
Advisory Hypothesis & Confidence Scoring

4. DETERMINISTIC RULES ENGINE
100% Scheme Validation — Rules Override AI

5. POLICY & SANCTIONS RE-SCREENING
Watchlist & Travel Rule Compliance Check

6. HUMAN REVIEW & MAKER-CHECKER
Hard $0 SoD Limit · Dual-Auth for ≥$250k

7. CONTROLLED EGRESS GATEWAY
Generates Schema-Compliant Repair ICD Contract

8. CLEARING DISPATCH (DISPATCHED)
Secure Transport ACK from Central Clearing Rail

9. FINAL SETTLEMENT (SETTLED)
pacs.002 ACCP or Internal DDA Ledger Posting

CROSS-CUTTING GOVERNANCE: Policy Rules | Maker-Checker ($0 SoD) | Dual-Auth (≥$250k) | SHA-256 Merkle Provenance | WORM 7-Yr Audit

CORE ARCHITECTURAL MANTRA: AI recommends. Controls validate. Humans authorize. Systems execute.

LIFECYCLE: EVENT ➔ INTELLIGENCE ➔ DETERMINISTIC CONTROL ➔ HUMAN AUTHORIZATION ➔ EXECUTION ➔ FINAL SETTLEMENT ➔ MERKLE AUDIT

PeX Fits Around the Bank

Non-intrusive integration. Bank-controlled data. Technology-stack neutrality.

NON-INTRUSIVE INTEGRATION: SURROUNDING THE BANK ESTATE

EXISTING BANK ESTATE (Core Banking · Payment Hubs · Clearing)
Finastra, ACI, FIS, Temenos, SAP, Thought Machine, SWIFT Alliance, Fedline

↓

STANDARDIZED ICD / ADAPTER LAYER
REST APIs | Enterprise MQ (IBM/MSMQ) | Event Streams (Kafka/Bus) | Batch SFTP

↓

READ-ONLY EVIDENCE
KYC & Sanctions
Client Master
Reference Data

PeX RESOLUTION CORE
41 Canonical Profiles
Deterministic Rules
HITL Maker-Checker

CONTROLLED EGRESS
Payment Gateways
DDA Posting
SWIFT / Clearing

↓

BANK-CONTROLLED EXECUTION & FINAL SETTLEMENT
PeX never moves funds directly — all repair messages execute via Bank Clearing Gateways.

TECHNOLOGY STACK NEUTRALITY
- *BANKING: Mainframe (z/OS), Distributed On-Prem, Private Cloud, Hybrid*
- *PAYMENTS: SWIFT MT (103/202), ISO 20022 MX (pacs/camt/pain), Fedwire, ACH*
- *INTEGRATION: REST JSON APIs, IBM MQ, Apache Kafka, SFTP Batch ICDs*
- *DEPLOYMENT: 100% Feature Parity on AWS, Google Cloud, Microsoft Azure, or On-Prem*

SECURITY & ZERO-PII DATA BOUNDARY

1. Bank Sensitive Data
Raw Names, IBANs, Tax IDs, Account Numbers

↓

2. Token Vault Enclave
Dedicated FIPS 140-2 Level 3 KMS CMK Isolation

↓

3. Surrogate Tokens (TKN_...)
PeX Core processes tokenized records exclusively

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4. Controlled Model Gateway
Advisory LLM prompts receive ZERO raw PII

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5. Approved AI Provider
AWS Bedrock · GCP Vertex AI · Azure OpenAI

DATA SOVEREIGNTY GUARANTEE
- **Bank payment data is NEVER used to train foundation models.**
- **Model retention governed strictly by bank tenant policies.**

ENTERPRISE INTEGRATION ► EXISTING BANK ESTATE ► STANDARDIZED ICDs & ADAPTERS ► PeX RESOLUTION CORE ► BANK-CONTROLLED EXECUTION

CORE TAKEAWAY: PeX adds a resolution capability around the bank's existing technology estate — without requiring technology standardization.

Why PeX Can Be Trusted and Scaled

Governed. Traceable. Reusable across heterogeneous bank environments.

THE PeX 4-LAYER ENTERPRISE TRUST STACK

1. EMBEDDED GOVERNANCE LAYER
Maker-Checker ($0 SoD) · Dual-Auth (≥$250k) · Pre-Execution Policy Guardrails

2. CRYPTOGRAPHIC TRACEABILITY LAYER
11-Stage SHA-256 Merkle Provenance Chain · 7-Year WORM Immutable Storage

3. COMMON RESOLUTION ONTOLOGY LAYER
Standardized ISO 20022 & BIAN Canonical Models · Cross-Silo Vocabulary

4. MULTI-ENTITY REUSABILITY LAYER
Reusable across Global Branches, Retail/Corporate Desks, & Subsidiary Hubs

REGULATOR READY & AUDIT-PROOF
"Every exception decision node is cryptographically sealed in an immutable Merkle chain for instant regulatory reconstruction."

ONTOLOGY TRANSFORMATION & REUSABILITY ACROSS BANKS

Bank A	Bank B	Bank C
Mainframe / MT	*Cloud / MX*	*Regional ACH*

PeX DOMAIN ONTOLOGY & CANONICAL MODEL
Normalizes disparate legacy representations into unified exception semantics.

41 CANONICAL EXCEPTION RESOLUTION PATTERNS
Common resolution intelligence deployed once, reused across all entities.

Global Subsidiary A	Commercial Line B	Correspondent Branch C
Instant Reuse	*Shared Rules*	*Consistent SLA*

THE FINAL CXO PROPOSITION
"PeX turns fragmented payment exceptions into a common, governed resolution capability — using AI where it adds intelligence, deterministic controls where they must remain authoritative, humans where accountability is required, and bank-controlled interfaces for execution."
Common resolution intelligence. Bank-controlled execution. Full traceability.

ENTERPRISE PROPOSITION ► COMMON INTELLIGENCE ► BANK-CONTROLLED DATA ► DETERMINISTIC GOVERNANCE ► HUMAN ACCOUNTABILITY ► FULL TRACEABILITY ► SCALABLE RESOLUTION

CORE TAKEAWAY: PeX turns fragmented exceptions into a common, governed resolution capability across the entire banking estate.